TEACHER GUIDE / SMALL BOX

Teaching with this workbook

Edition 1.1 incorporates Steve's 10 September 2026 instruction: "the lid on the small box can be optional". The student PDF has 50 A4 pages for 25 double-sided sheets. Native Word pagination needs separate verification. This guide supplies answers and evidence expectations, not assessment marks or a new practical operating procedure.

Two valid outcomes

Students may choose an open box or the lidded version. All nine named sections remain taught across seventeen readings. All 36 MCQs and twelve written prompts match the current source wording, including the amended Module 1 and 3 questions. Conditional lid scenarios teach knowledge; they do not assign lid construction. Twelve folio pages accept evidence for the actual choice. Do not require a lid, hinge or lid-chamfer task from students choosing the open option.

Teach with the source hub

Use the three module slide decks and the course-selected videos for an observation or discussion purpose. The decks support section focus, checking and evidence, and do not contain dimensional construction instructions. Use the hierarchy-of-controls video with page 8, face-edge/face-side with page 7, glue with page 23 and finishing with pages 36–37. Current teacher demonstrations, SOPs and approved product information govern practical work.

Resolve the chosen technical source

The frozen source describes the original lidded pathway but contains no dimensional plan PDF. The dated teacher-amendment.json takes precedence for lid optionality. Record the chosen option and its teacher-confirmed component layout on page 3. Stock, dimensions, joint details, setup, controls, products and any chosen hinge fitting remain teacher/source controlled. No divider or revised joint arrangement follows automatically from the amendment.

Evidence and assessment

The folio accepts the student's own build photograph, sketch or check. The current program names checkpoint photographs in periods 5, 10 and 15; retain these as directed. The pair discussion on page 12 supports its authorised collaboration requirement. Agree evidence attachment, backup and submission. Browser storage is a working copy, not automatic submission. Formal marks, dates, conditions and adjustments come from the teacher-issued assessment.

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Source boundaries and illustrations

Original source snapshot: https://stevencowell.github.io/Small-Box/ at commit 41c3d8dd77dc4cd5b6974c50babec6a15a648d9f. The 10 September teacher amendment and copied Module 1 and 3 lesson-data.js updates take precedence for current questions and lid optionality. Module 2 questions are unchanged. The original snapshot remains intact. Theory sections, folio, teacher v2.1 program and three presentation decks supply the underlying learning map.

The teacher amendment makes the lid optional. The original source remains unchanged as evidence of its lidded version. The old hero and box.png are no longer unsuitable merely because they are open; they remain excluded because their divider is not automatically approved. The dormant divider card is already suppressed by the source script. The new open-box illustration has no divider.

The advanced suggestion of adding a lid can now be a future option for an open box, with technical/teacher confirmation. It is not compulsory. The page 50 task accepts another justified refinement to the chosen version. Adhesive and finish comparisons still require actual product information; the amendment supplies no product guarantees, dimensions, stock assignments or machine instructions.

Three generated photographs show optional open and lidded examples and a material study. Fifteen labelled vectors explain source, reference, control, feature, fit and evidence relationships. The sequence marks lid stages as optional; lid/chamfer/hinge figures are conditional examples. These are teaching aids, not approved construction plans, machinery setups or student work. Visible counts, colours and proportions create no specifications.

The source finish-folio prompt mentions checks between coats. Preserve its wording but accept not applicable where the approved process has no between-coat stage; require the actual readiness or final check instead. Similarly, a risk pathway is not a claim that an incident happened. Accept honest evidence gaps, planned work labelled as planned, equivalent explanations and teacher-confirmed alternatives. Completing a page is never practical sign-off.

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Applied teaching and discussion

Module 1: see the decision chain

Record the chosen option before stock/layout decisions. Q1–3 use the current optional-lid brief; Q3 asks what would be required if hinges were selected. Everyone can answer that conditional knowledge question, then record the component and authority needs of their actual choice. Separate visible form from specifications. The material study and datum diagram teach observation and references, not numeric values.

Module 2: make a check useful

The feature gallery shows flared dovetail, rectangular finger and rebate step concepts. The sequence follows the confirmed order without inventing intermediate machining instructions. During dry fit, demand contact, alignment and demonstrated squareness evidence. In the diagnosis example, one corner not seating is an observation, while orientation/contact remain possible causes. Accept multiple plausible causes only when accompanied by a suitable teacher-supported investigation.

Extension: refine a design responsibly

Link the adhesive comparison on page 29 to the future sketch on page 50. Students may refine their actual open or lidded option, including considering a future optional lid for an open box. Require benefit, consequence and technical/teacher confirmation before testing. A divider or another additional feature is not automatically authorised.

Module 3: knowledge and practical choice

All twelve current Module 3 checks remain useful. Q26–28 and Q33 explicitly ask about a selected lidded or hinged version; these are knowledge scenarios. Q32 and Q36 now cover the features applicable to either actual box. The current written prompt on page 43 contains two explicit pathways. Open students explain opening, exposed-edge and access checks. Lidded students explain relevant alignment, movement and approved hinge controls. Neither response invents work they did not do.

PMI and transfer

Require a supported strength, a specific limitation and a useful next question. The advanced task adds a second strength and improvement. A stronger inspection routine can be a more worthwhile improvement than decoration. Each future habit needs a trigger, a reason from the Small Box and a check that could show whether it helped. The next unit's own authority still controls its construction.

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Pages 3–12: answers and evidence

Student page 3: Begin with the confirmed brief

Record the chosen open-box or lidded option, current teacher-issued technical resource and its location/version. Confirm the component layout for that option. Pending technical detail must be resolved before dependent practical work; optional lid does not authorise a divider, new joint arrangement or guessed values.

Student page 10: Check the chosen brief

1. A. Correct. The approved project is a small timber box with an optional lid.

2. A. Correct. Include the top/lid stock only when that feature is selected.

3. B. Knowledge scenario about a selected lid; it does not require every student to make one. Correct. Hinges are teacher supplied or teacher approved.

Choose either valid option under the dated amendment. Identify the actual component list from the teacher-approved technical source; no unused lid components or hinges required for an open box. Dimensions remain source/teacher controlled; hardware if needed remains teacher supplied/approved. No automatic divider or new joinery authority.

Student page 11: Check the mark

4. A. Correct. These sources control the project detail.

5. A. Correct. This creates a controlled and checkable layout.

6. B. Correct. Stop and resolve the uncertainty first.

Printed order: 4, 1, 5, 2, 3. Identify approved information; establish the reference; mark through the demonstrated method; check orientation and position; obtain the required pre-cut confirmation. Accept discussion that teacher checks may also occur earlier, but none is omitted before cutting.

Student page 12: Check the controls

7. A. Correct. These checks support reliable planning.

8. A. Correct. This shows a traceable process.

9. A. Correct. Stronger controls should be considered before relying on PPE alone.

Confirm demonstrated secure support/clamping; stop and confirm required extraction/dust controls and PPE through teacher/SOP; stop and obtain instruction/permission for unclear next action. Do not remove guards, modify equipment or treat PPE as the whole solution.

Teacher-authorised collaboration: students compare a real hazard/control choice, identify meaningful feedback and a suitable teacher/SOP check. They do not authorise one another's machine use. Accept a revised or retained choice when justified.

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Pages 13–14: answers and evidence

Student page 13: Check your authority

10. B. Correct. Practical work remains teacher controlled.

11. B. Correct. Stop, reassess and control the hazard.

12. A. Correct. This is a sound starting point for the approved project.

The video supports conceptual understanding or an observation focus. It does not replace current school SOP, teacher demonstration, required controls/supervision or explicit permission. Student must wait for the required practical instruction and authority; confidence or a completed quiz does not remove this boundary.

Student page 14: Explain the brief and marks

Expected current response: choose an open timber box or a lidded option with the teacher. Components/stock match the chosen approved plan: sides and bottom, plus top/lid only if selected. Hinges are teacher supplied/approved only if a hinged lid is chosen. No lid stock, hardware or divider is compulsory for the open choice. Accept equivalent source-grounded wording and actual evidence; do not require the model verbatim or invent completed practical results.

Expected response: I will begin with the approved teacher-issued source and mark from a reliable reference face and edge. I will verify each mark before cutting. If a value or orientation is unclear, I will stop and ask the teacher rather than guess. Also check the source concepts: source, marking, verification. Accept equivalent source-grounded wording and actual evidence; do not require the model verbatim or invent completed practical results.

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Pages 15–17: answers and evidence

Student page 15: Explain safe preparation

Expected response: A moving or unsecured workpiece is a hazard. Before relying on PPE, the work should be secured and the approved guard or other engineering control used. The teacher demonstration and current school SOP control the exact setup and permission to proceed. Also check the source concepts: hazard, control, authority. Accept equivalent source-grounded wording and actual evidence; do not require the model verbatim or invent completed practical results.

Expected response: I will save a dated source check, a photo of verified marks, a note recording the required teacher check and a brief record of any correction. Together these items will show what I checked, what changed and why before practical work continued. Also check the source concepts: evidence, sequence, decision. Accept equivalent source-grounded wording and actual evidence; do not require the model verbatim or invent completed practical results.

Student page 16: Criteria you can check

Open option: opening-edge/interior-access evidence. Lidded option: relevant alignment/movement/hardware evidence. Both: body joint contact, square/stability and surface/finish checks against the approved plan. Avoid invented dimensions, marks or completed quality claims.

Name the valid chosen option and two specific requirement/check links. Actual stock follows that option's technical source; hinges only for a chosen hinged lid. Open box is a completed design option, not a failed or unfinished lid task.

Identify authentic own-build evidence and caption the stage, relevant observation/decision and result. Accept pending status where appropriate; never claim workbook/generated imagery as student practical evidence.

Student page 17: A layout that stays readable

Own approved layout with intelligible references, identities/orientation and actual source reference. Do not infer dimensions, counts or tolerances from workbook illustrations. A proposed layout must be labelled planned and checked before cutting.

Explain the sequence from approved source to consistent reference, clear mark, orientation check and required pre-cut confirmation. Link it to avoiding irreversible error; actual evidence or an honestly planned check is acceptable.

Identify authentic own-build evidence and caption the stage, relevant observation/decision and result. Accept pending status where appropriate; never claim workbook/generated imagery as student practical evidence.

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Pages 18–27: answers and evidence

Student page 18: Make the control visible

Three actual relevant tasks/hazards with suitable teacher/SOP controls and permission points. Consider stronger controls before PPE. Do not invent machine setups or an incident, and do not treat hazard recognition as practical sign-off.

Name the hazard pathway and strongest relevant approved control, explain how it reduced the possibility of harm and record the actual check. The source wording asks what prevented the pathway; accept a risk explanation without claiming an incident occurred or that all risk disappeared.

Identify authentic own-build evidence and caption the stage, relevant observation/decision and result. Accept pending status where appropriate; never claim workbook/generated imagery as student practical evidence.

Student page 19: Inspect and place the timber

Actual material inspection and source-bound layout. Show the location of any observed feature and reasoned placement; consult teacher for suitability. If no change needed, record inspected/retained with reasons. No invented species, defects, stock dimensions or allocations.

Connect observed grain/defects/straightness/suitability to a placement decision and responsible material use. If inspection confirmed the original layout, explain that; no artificial change is required.

Identify authentic own-build evidence and caption the stage, relevant observation/decision and result. Accept pending status where appropriate; never claim workbook/generated imagery as student practical evidence.

Student page 26: Check the features

13. A. Correct. Dovetails are part of the approved Small Box sequence.

14. A. Correct. A rebate helps house or locate another component.

15. A. Correct. Both are named in the approved pathway.

Rebate: recessed step locating the intended component and its contact/alignment. Joint: intended interlocking mating parts meet in the approved orientation. Exact locations and geometry must be verified against the approved source; no invented dimensions or number of features.

Student page 27: Check the dry fit

16. A. Correct. The finger joint is confirmed in the sequence.

17. A. Correct. Dry fitting exposes issues while they can still be corrected.

18. A. Correct. Square glue-up protects the whole assembly.

Verify source/orientation and joint contact; apply demonstrated whole-box squareness/alignment check; identify intended component/rebate relationship against source and teacher direction. Do not force, add adhesive to hide gaps or remove material randomly. Record the observation and obtain authorised correction.

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Pages 28–30: answers and evidence

Student page 28: Check the correction

19. B. Correct. Controlled correction protects accuracy and evidence.

20. A. Correct. This makes the correction traceable.

21. A. Correct. This documents the full decision.

Several changes obscure which caused the result and may introduce new faults. Stop, verify source/orientation and observation, ask teacher to approve one controlled change, repeat the relevant dry-fit/contact/alignment/square check, and record before/action/after. Do not prescribe a particular removal or pressure without actual diagnosis.

Student page 29: Check the evidence

22. A. Correct. These sources control practical work.

23. A. Correct. This follows a controlled high-level sequence.

24. A. Correct. This closes the joinery and assembly learning loop.

Use actual teacher-provided labels or product information for two adhesive examples (e.g. products within PVA, polyurethane or epoxy families). Record supported intended use and a control/timing requirement with product identity. If unavailable write information required. No blanket waterproof/gap-filling claim or unauthorised product change.

Student page 30: Explain sequence and fit

Expected response: The approved sequence prepares the dovetail corner joints before the back rebate. It then includes the finger joint and the bottom rebate. The exact dimensions, methods and settings come from the approved project resources and teacher demonstration. Also check the source concepts: dovetails, rebates, finger joint. Accept equivalent source-grounded wording and actual evidence; do not require the model verbatim or invent completed practical results.

Expected response: During the dry fit I would check that the confirmed joints seat consistently and that the components align without forcing. I would verify the box is square and record any correction. Glue-up would begin only after the required teacher check and approved process. Also check the source concepts: fit, alignment, authority. Accept equivalent source-grounded wording and actual evidence; do not require the model verbatim or invent completed practical results.

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Pages 31–33: answers and evidence

Student page 31: Explain a controlled correction

Expected response: The dry fit showed that one corner did not seat consistently and the box was not square. I stopped, identified the affected fit and made one teacher-approved correction. I then repeated the dry fit and recorded the new square and contact checks. Also check the source concepts: issue, correction, recheck. Accept equivalent source-grounded wording and actual evidence; do not require the model verbatim or invent completed practical results.

Expected response: I would slow down during the dry fit because this is the last reversible point before glue-up. Checking joint contact and squareness protects the finished assembly. I would record the issue, correction and recheck with notes and photographs. Also check the source concepts: control, quality, evidence. Accept equivalent source-grounded wording and actual evidence; do not require the model verbatim or invent completed practical results.

Student page 32: Record the decisive dry fit

Record actual identified approved joint/rebate relationships and relevant contact/alignment/square observations, teacher-supported correction if needed and repeated check. An uncompleted stage is pending, never invented. Exact feature arrangement comes from technical source.

Select a real joint/alignment check and explain why its evidence increased or changed confidence. Describe the initial condition, any controlled correction and the repeated check. More clamp force is not evidence that a faulty dry fit is solved.

Identify authentic own-build evidence and caption the stage, relevant observation/decision and result. Accept pending status where appropriate; never claim workbook/generated imagery as student practical evidence.

Student page 33: Keep the assembly controlled

Source-bound record of actual teacher-demonstrated arrangement, secure support and relevant checks. Do not prescribe an arrangement based on the conceptual workbook figure or invent pressures, quantities or curing times.

Describe the actual teacher-approved holding/support arrangement and staged checks that controlled movement. Explain the relevant critical moment and observed result. If the assembly has not occurred, clearly label the plan and update after the real check.

Identify authentic own-build evidence and caption the stage, relevant observation/decision and result. Accept pending status where appropriate; never claim workbook/generated imagery as student practical evidence.

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Pages 39–41: answers and evidence

Student page 39: Check your chosen option

25. A. Correct. A chamfer is a bevelled edge.

26. A. Knowledge scenario about a selected lid; it does not require every student to make one. Correct. Lid chamfers apply when the lid is selected and specified in the approved source.

27. A. Knowledge scenario about a selected lid; it does not require every student to make one. Correct. The teacher controls hinge suitability.

Lidded option: stop forcing; teacher reviews approved hardware, body/lid alignment, placement/method; record actual movement check, approved correction and recheck. Open option: locate the rough opening edge, confirm approved correction while preserving form, then reinspect opening/interior and record before/action/result. Appearance alone proves neither; no guessed removal values or machine steps.

Student page 40: Check surface readiness

28. A. Knowledge scenario about a selected lid; it does not require every student to make one. Correct. These checks protect function and quality.

29. A. Correct. These are sound readiness checks.

30. A. Correct. These sources control finish use.

Printed order: 3, 1, 4, 2. Inspect complete box; identify/location-record issues; use approved correction/preparation and dust controls; inspect again and confirm finish readiness. Surface/product/control confirmation precedes actual finish application.

Student page 41: Check finish decisions

31. A. Correct. Finishing remains a controlled process.

32. A. Correct. This gives a balanced product check.

33. A. Knowledge scenario about a selected lid; it does not require every student to make one. Correct. This demonstrates the full control sequence.

Use actual teacher-provided product information, not general family guarantees. Record precise supported appearance/duty and controls/waste/care evidence. Accept information required if data is unavailable. Environmental conclusions must link actual handling, quantity, waste or durability; no product is assigned by this comparison.

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Pages 42–43: answers and evidence

Student page 42: Check your judgement

34. A. Correct. Evidence makes the evaluation credible.

35. A. Correct. This supports learning and credibility.

36. A. Correct. This closes the project learning loop.

Require a strength, limitation/effect and future question supported by relevant evidence. Open option might use checked opening edges/access or body stability; lidded option may use actual movement/hinge checks. Both can discuss joint/finish quality and systematic inspection. Do not require lid evidence from an open box or accept invented own-work results.

Student page 43: Explain readiness for your option

Choose the applicable branch. Lidded option: verify approved lid alignment, intended movement without force, teacher-approved hinges, placement/method and required teacher check before fitting. Open option: explain opening-edge and interior-access inspection plus body joint/stability checks, unresolved faults and any teacher-supported correction/recheck. No lid or hardware evidence is required from open-box students. Accept equivalent source-grounded wording and actual evidence; do not require the model verbatim or invent completed practical results.

Expected response: The box is ready when the full surface is clean, dry and even, with no unresolved marks or fit problems. I would correct any remaining issue before finishing. The exact product and process come from teacher direction, the school SOP and approved product information. Also check the source concepts: surface, defects, authority. Accept equivalent source-grounded wording and actual evidence; do not require the model verbatim or invent completed practical results.

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Pages 44–46: answers and evidence

Student page 44: Evaluate and transfer

PMI for the actual choice: both versions use joint condition, stability and surface/finish evidence. Open-box evidence includes opening edges/interior access; lidded evidence may add movement/hinge security. Require a supported plus, honest minus and worthwhile next question. The source model is an example, not a claim about the student's own work. Accept equivalent source-grounded wording and actual evidence; do not require the model verbatim or invent completed practical results.

Expected response: In my next project I will verify the approved source before marking and use a dry fit before irreversible assembly. I will also record each correction and teacher check. These habits improve accuracy, safety and the quality of my evidence. Also check the source concepts: transfer, habit, benefit. Accept equivalent source-grounded wording and actual evidence; do not require the model verbatim or invent completed practical results.

Student page 45: Prove surface readiness

Inspect the actual option: opening/interior and body, or a chosen lid's relevant features. Use demonstrated preparation/dust controls and record issue/correction/recheck. No compulsory lid preparation, invented grit sequence or automatic readiness.

Evidence should support clean, dry, even and sufficiently prepared condition for the actual approved finish. Include faces/edges rather than one smooth patch, dust/handling controls and recheck of faults. Student should distinguish surface judgement from required teacher authorisation.

Identify authentic own-build evidence and caption the stage, relevant observation/decision and result. Accept pending status where appropriate; never claim workbook/generated imagery as student practical evidence.

Student page 46: Record the actual finish process

Name actual approved product/source, required controls and only real process stages. Include timing/coat observations only when product/teacher supplies them. Accept not applicable for between-coat stage; no extra coats, schedules or waste procedure invented.

Relate an actual interim check to a quality improvement, with observation, authorised action and result. If no interim/between-coat check was required, explain that honestly and describe actual readiness/final check. The original prompt does not authorise multiple coats.

Identify authentic own-build evidence and caption the stage, relevant observation/decision and result. Accept pending status where appropriate; never claim workbook/generated imagery as student practical evidence.

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Pages 47–49: answers and evidence

Student page 47: Build a clear evidence flow

Authentic chronological own-build evidence for the chosen option. Open students show relevant opening/body evidence; lid records only if chosen. Identify pending/gaps honestly. Do not submit generated illustrations as own work or expose unrelated private information.

Select a meaningful design/process decision and a genuine evidence item that directly supports it. Caption stage/action/reason/result; avoid claims beyond what the image/check can demonstrate. Browser working data needs teacher-directed backup/submission.

Identify authentic own-build evidence and caption the stage, relevant observation/decision and result. Accept pending status where appropriate; never claim workbook/generated imagery as student practical evidence.

Student page 48: Explain one worthwhile correction

One real issue, supported diagnosis, teacher-approved correction and repeated relevant check with honest outcome. If no actual correction yet, use a clearly labelled planned investigation and return to record what happens; do not manufacture a fault.

Accurate own-work or clearly planned before/after views; same relevant check, controlled change and observed result. No new technical values or unapproved methods.

Explain why this real correction mattered to learning: evidence, cause, authorised action, recheck and transferable lesson. Accept an unsuccessful first test when honestly evaluated. Distinguish observation from a possible cause.

Identify authentic own-build evidence and caption the stage, relevant observation/decision and result. Accept pending status where appropriate; never claim workbook/generated imagery as student practical evidence.

Student page 49: A balanced final judgement

Criterion-linked observable plus; honest limitation/effect; worthwhile question or realistic next improvement. Include function, fit or finish evidence from actual checks. Extension can add second strength/improvement within response. No invented tests or performance guarantees.

Judge the chosen brief using joint condition/stability and finish. Add opening-edge/interior-access evidence for an open box, or lid movement/hinge security for the lidded option. Use real evidence and relevant limitations/improvements. An intentionally open box is not unfinished because it lacks a lid.

Identify authentic own-build evidence and caption the stage, relevant observation/decision and result. Accept pending status where appropriate; never claim workbook/generated imagery as student practical evidence.

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Page 50: answers and evidence

Student page 50: Carry a better routine forward

Two specific transferable routines supported by actual learning, such as verify source/reference before mark; complete dry fit before glue; one-change/recheck correction; systematic surface inspection; accurate evidence chronology. Include trigger and measurable observation rather than vague “try harder”.

A justified future refinement to the actual open or lidded choice. A future optional lid can be considered for an open box, but is not compulsory; a divider/other new feature still needs approval. Explain benefit/consequence/technical needs without inventing values or changing the current build automatically.

Explain two routines with actual Small Box evidence and why each will support the next project. One may preserve a strength; one may address a limitation. The future project’s own plan and teacher procedures must still control practical details.

Identify authentic own-build evidence and caption the stage, relevant observation/decision and result. Accept pending status where appropriate; never claim workbook/generated imagery as student practical evidence.